

# Do you Want to Meet at Coldstone?



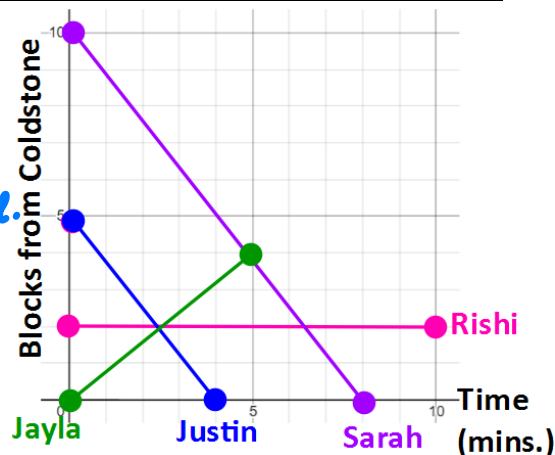
After analyzing the prices of Coldstone ice cream sizes, Sarah decides she's "Gotta Have it" and makes a plan to meet a few friends there for a treat. She calls Jayla, Justin, and Rishi. Jayla was already there and had to leave, Justin was 5 blocks away, and Rishi was 2 blocks away.

1. Explain the relationship between time and blocks from Coldstone for each person.

As time increases, Jayla's distance from coldstone increases, Justin's & Sarah's decreases & Rishi stays the same.

2. Write an equation in slope-intercept form for Justin's distance from Coldstone.

$$y = -\frac{5}{4}x + 5$$



3. Write an equation in slope-intercept form for Sarah's distance from Coldstone.

$$y = -\frac{5}{4}x + 10$$

4. Identify one similarity and one difference between Justin and Sarah's equations. What do each of these mean in the context of the problem?

parallel lines: same slope, different y-intercepts

They have the same slope, but different y-int. They are moving at the same speed, but started at different locations.

5. Write an equation in slope intercept form for Jayla's distance from Coldstone. Is it accurate to say that Jayla biked the same speed but in the opposite direction as Sarah and Justin?

perpendicular lines: opposite reciprocal slopes

$y = \frac{4}{5}x$  No, she biked 4 blocks in 5 minutes, not 5 blocks in 4 min like Sarah & Justin

6. Poor Rishi is stuck in a traffic jam. How can you tell this from the graph?

no rise slope = 0

his position is always 2 blocks from coldstone

7. Write an equation for Rishi's distance from Coldstone.

$$y = 2$$

8. Jayla and Justin are the same distance from Coldstone at the same time. Show algebraically at what time this occurs.

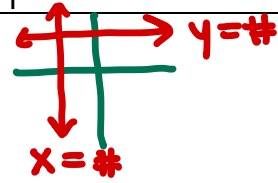
$$\underbrace{\left(-\frac{5}{4}x + 5\right)}_{\text{Justin}} = \underbrace{\left(\frac{4}{5}x\right)}_{\text{Jayla}} \quad 20$$

$$\begin{aligned} -25x + 100 &= 16x \\ 100 &= 41x \\ x &= \frac{100}{41} = 2.4 \text{ mins.} \end{aligned}$$

## Lesson 0.6 – Reasoning with Slope

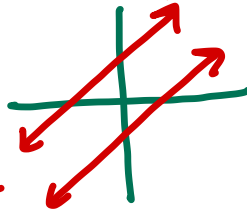
QuickNotes

Horizontal line: a line w/ no rise  $\rightarrow$  slope = 0  
 Vertical line: a line w/ no run  $\rightarrow$  slope = undef.



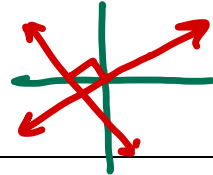
Parallel lines

- never intersect
- same slope
- different y-intercept



Perpendicular lines

- cross at  $90^\circ$
- opposite reciprocal slopes:  $m_1 = \frac{a}{b}, m_2 = -\frac{b}{a}$



### Check Your Understanding

- A line with a slope of  $\frac{1}{2}$  passes through the point  $(0, -3)$ .
  - Write an equation of a line parallel to this one.

$$y = \frac{1}{2}x + 5 \text{ (answers will vary)}$$

- Write an equation of a line perpendicular to this one with the same y-intercept.

$$y = -2x - 3$$

- Line  $m$  passes through the points  $(-1, 4)$  and  $(5, k)$ . The line given by  $y - 7 = -\frac{3}{2}(x + 4)$  is perpendicular to line  $m$ . Find the value of  $k$ .

Line  $m$  has slope of  $\frac{2}{3}$

$$\frac{\Delta y}{\Delta x} = \frac{2}{3} \quad \Delta x = 6 \quad \Delta y = 4 \quad k = 4 + 4 = \boxed{8}$$

- Write the equation of the vertical line passing through the point  $(2, 3)$ .

$$x = 2$$

- Write the equation of the horizontal line passing through the point  $(2, 3)$ .

$$y = 3$$

- Consider the line given by  $5x - 4y = 20$ . Which of the following equations is perpendicular to that line?

- $4x - 5y = 20$
- $-4x - 5y = 20$
- $-\frac{1}{5}x - 4y = 20$
- $10x - 8y = 40$

$$\text{Slope} = \frac{-5}{-4} = \frac{5}{4}$$

$$\text{opp. reciprocal} = -\frac{4}{5}$$